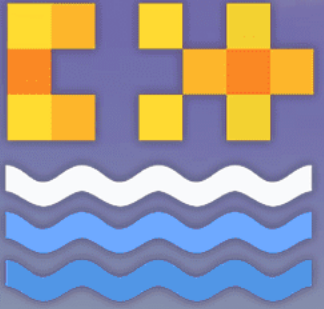




A New Dragon in the Den

Fast Conversion From
Floating-Point Numbers

Cassio Neri

2024

Hercules slaying the Hydra



Hercules slaying the Hydra

Louis Chéron

Oil painting, ca. 1690-1725

Victoria & Albert Museum, London, UK

Hercules slaying the Hydra



Hercules slaying the Hydra
Louis Chéron
Oil painting, ca. 1690-1725
Victoria & Albert Museum, London, UK

Bicho de sete cabeças.

Seven-headed beast.

Hercules slaying the Hydra



Hercules slaying the Hydra
Louis Chéron
Oil painting, ca. 1690-1725
Victoria & Albert Museum, London, UK

Bicho de sete cabeças.
Seven-headed beast.



Medieval French Apocalypse Tapestry, Produced between 1377 and 1382.
Château d'Angers, France
Credits: Jean-Pierre Dalbéra, CC BY 2.0, via Wikimedia Commons.



Album The Number of the Beast
Iron Maiden, 1982.

Scream for me
Folkestone !!!



Bruce Dickinson in Brno, Czech Republic, June 8, 2014
Credits: Vaclav Salek/CTK Photo/Alamy Live News

Hercules slaying the Hydra



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Louis Chéron
Oil painting, ca. 1690-1725
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Bicho de sete cabeças.

Seven-headed beast.

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Bicho de sete cabeças.

Seven-headed beast.

Não é um bicho de sete cabeças.

It's not a seven-headed beast.

Hercules slaying the Hydra



Hercules slaying the Hydra
Louis Chéron
Oil painting, ca. 1690-1725
Victoria & Albert Museum, London, UK

Bicho de sete cabeças.

Seven-headed beast.

Não é um bicho de sete cabeças.

It's not a seven-headed beast.

It's not rocket science.

A New Dragon in the Den

Fast conversion from floating-point numbers

Cassio Neri
(Independent Researcher)

FGV EMAp - Rio de Janeiro





<https://godbolt.org/z/a8qETqzbj>

```
string s3 = (stringstream{} << pi).str();           // 3.14159

string s4 = to_string(pi);                           // 3.141593

string s5(32, '\\0');
auto r5 = to_chars(s5.data(), s5.data() + s5.size(), pi); // 3.141592653589793
s5.resize(r5.ptr - s5.data());

string s6 = format("{} ", pi);                       // 3.141592653589793
```

π



<https://godbolt.org/z/a8qETqzbj>

```
string s3 = (stringstream{} << pi).str();           // 3.14159

string s4 = to_string(pi);                           // 3.141593

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.NET

<https://godbolt.org/z/czhrT1hhh>



<https://godbolt.org/z/qh4zdqG6>



<https://godbolt.org/z/8M483Kvba>

JS

<https://godbolt.org/z/nG88rv5f>



<https://godbolt.org/z/5aseWr7fz>



<https://godbolt.org/z/Y46WY4ooY>



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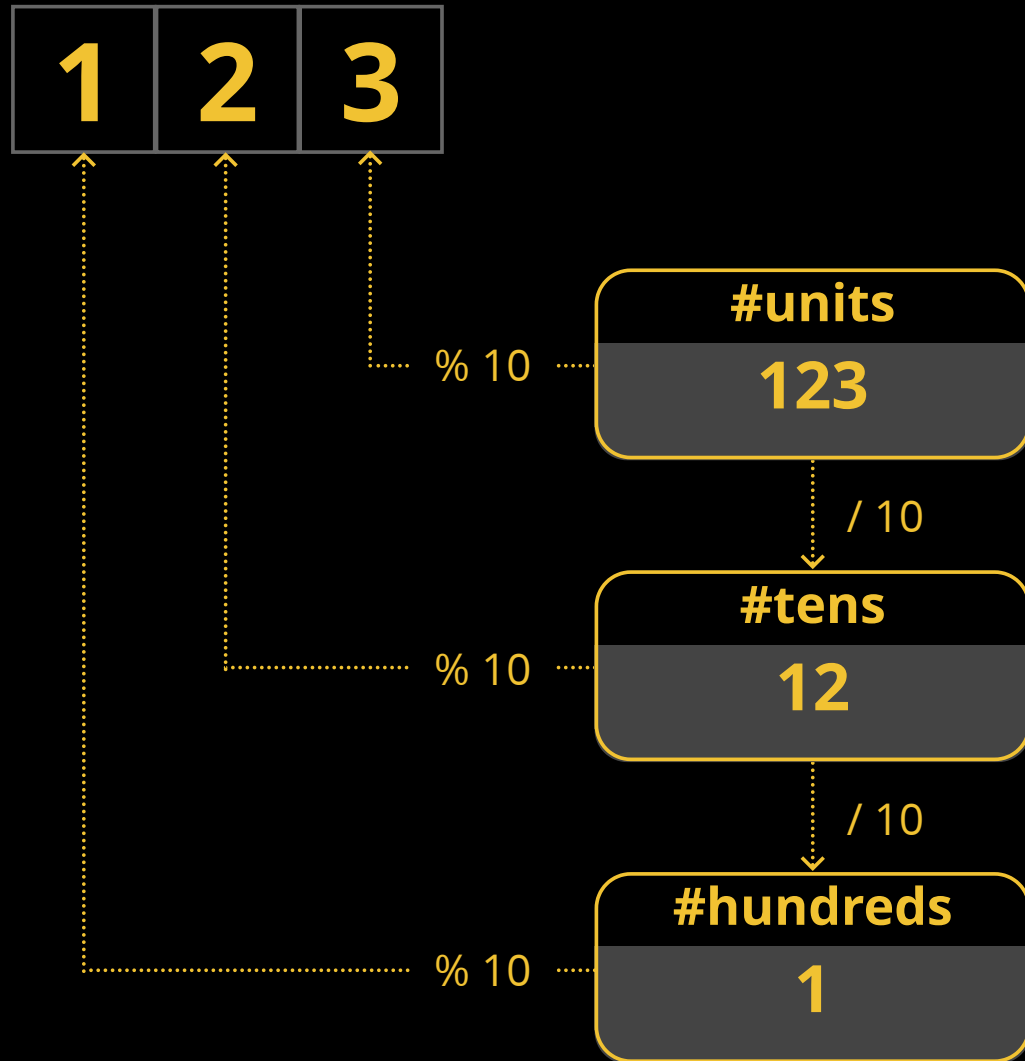


<https://godbolt.org/z/5aseWr7fz>

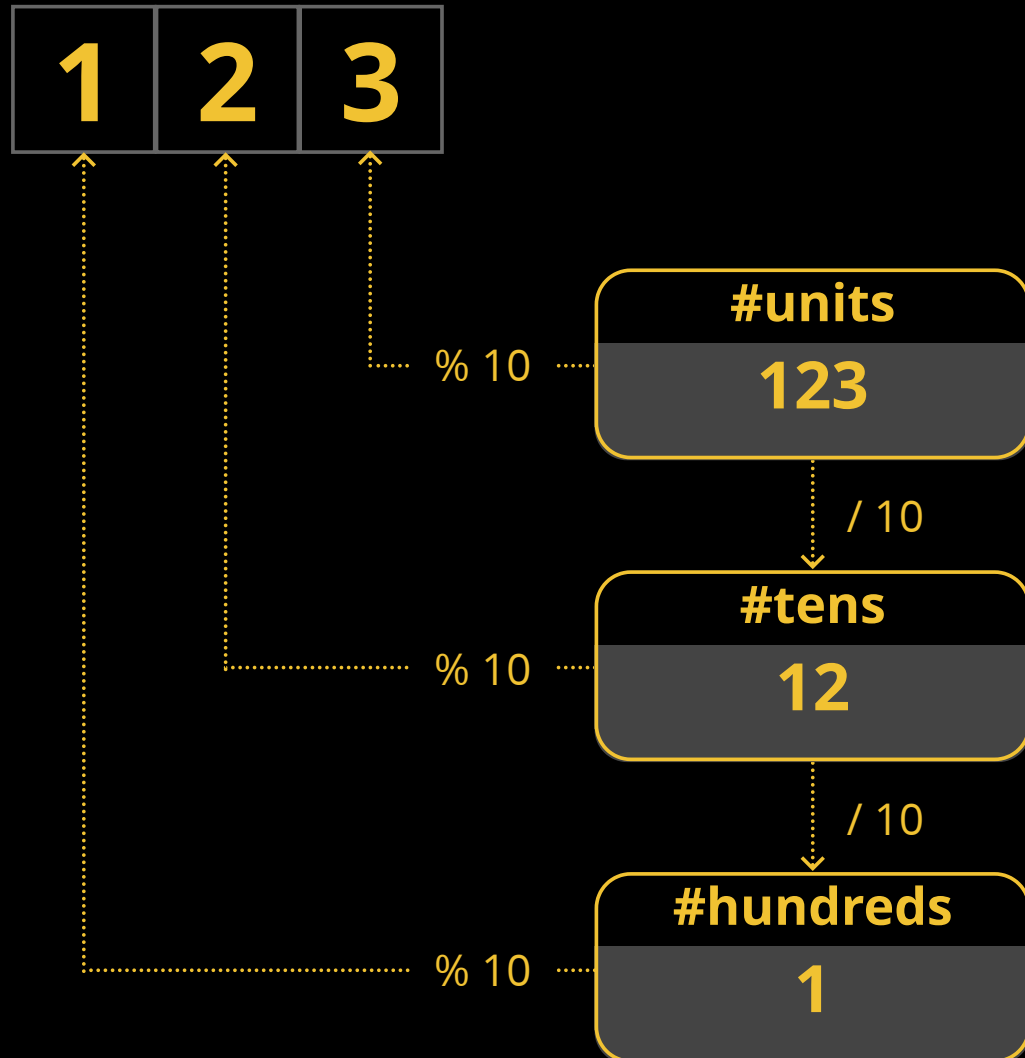


<https://godbolt.org/z/Y46WY4ooY>

Integer to string



Integer to string



<https://godbolt.org/z/1G9hbKMnP>

```
string convert(unsigned int n) {  
  
    size_t size = number_of_digits(n);  
    string str(size, '\\0');  
    char* p = &str.back();  
  
    do {  
  
        *p = n % 10 + '0';  
        n /= 10;  
  
        --p;  
  
    } while(n);  
  
    return str;  
}
```

(149 digits after the dot)



$$2^{-149} = 0.000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,001\,401\,298\,464\,324\,817\,070\,923\,729\,583\,289\,916\,131\,280\,261\,941\,876\,515\,771\,757\,068\,283\,889\,791\,082\,685\,860\,601\,486\,638\,188\,362\,121\,582\,031\,25$$

(149 digits after the dot)



$$\approx 10^{-45}$$



Decimal fixed-point representation

1234

Decimal fixed-point representation

12.34

Decimal floating-point representation

exponent
1
mantissa
10¹ x 2.34

Decimal floating-point representation

exponent
 $10^1 \times 2.34$
mantissa

0,0 0 0,0 0 0,0 2 3.4 0

Decimal floating-point representation

exponent
 $10^1 \times 2.34$
mantissa

0,0 0 0,0 0 0,0 2 3.4 0 

Decimal floating-point representation

exponent $\neq 0$



mantissa's first digit $\neq 0$
(scientific notation)

exponent
 $10^1 \times 2.34$
mantissa

0,0 0 0,0 0 0,0 2 3.4 0

Decimal floating-point representation

exponent

mantissa

$$10^9 \times 9.99 \times 10^{-3}$$

9,9 9 0,0 0 0.0 0 0,0 0

Decimal floating-point representation

biased exponent $\neq 0$
 $\implies$
mantissa's first digit $\neq 0$
(scientific notation)

$$10^{\text{biased exponent } 9-3} \times \text{mantissa } 9.99$$

9,9 9 0,0 0 0.0 0 0,0 0

Binary floating-point representation

Exponent and mantissa are in binary.

$$2^{000-11} \times 0.00$$

biased exponent

mantissa



Binary floating-point representation

Exponent and mantissa are in binary.

biased exponent $\neq 0$



mantissa's first bit = 1
(normal)

biased exponent
000-11
2 **x 0.000**
mantissa



Binary floating-point representation

Exponent and mantissa are in binary.

biased exponent $\neq 0$



mantissa's first bit = 1
(normal)

biased exponent
000-11
2 **x 0.000**
mantissa

—— Centred value

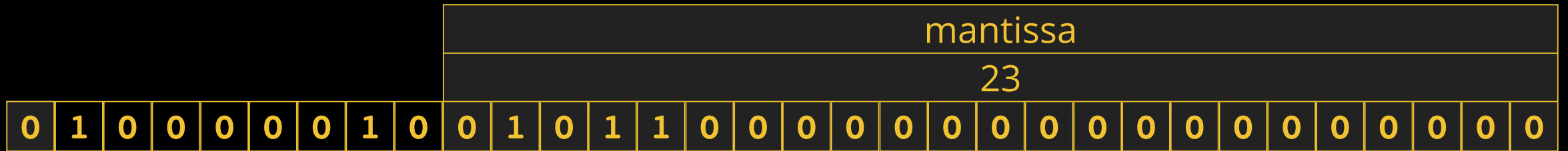
..... Uncentred value



IEEE-754 representation

- *Binary32* (`float` and `std::float32_t`)
- *Binary64* (`double` and `std::float64_t`)
- *Binary128* (`__float128` and `std::float128_t`)
- Others

Binary32



[illegible]

Binary32

[illegible]

Binary32

±	biased exponent								mantissa																				
1	8								23																				
0	1	0	0	0	0	0	1	0	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

0 sign: + (0) or – (1)

Binary32

±	biased exponent								mantissa																				
1	8								23																				
0	1	0	0	0	0	0	1	0	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

+

Binary32

±	biased exponent								mantissa																				
1	8								23																				
0	1	0	0	0	0	0	1	0	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

1 0 0 0 0 0 1 0

biased exponent $\neq 0 \implies$ normal

- 0 1 1 1 1 1 1 1

bias: **01111111** (127, normal) or **011111110** (126, subnormal)

0 0 0 0 0 0 1 1

+

Binary32

±	biased exponent								mantissa																				
1	8								23																				
0	1	0	0	0	0	0	1	0	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

biased exponent $\neq 0 \implies$ normal

+2⁰⁰⁰⁰⁰⁰¹¹x1.

Binary32

±	biased exponent								mantissa																				
1	8								23																				
0	1	0	0	0	0	0	1	0	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

$+2^{00000011} \times 1.010110000000000000000000$

Binary32

±	biased exponent								mantissa																				
1	8								23																				
0	1	0	0	0	0	0	1	0	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

$+2^{00000011} \times 1.010110000000000000000000$

$+2^3 \times 1.34375$

Binary32

±	biased exponent								mantissa																				
1	8								23																				
0	1	0	0	0	0	0	1	0	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

$+2^{-00010100} \times 101011000000000000000000$

$(+2^{-20}) \times 11,272,192.343,75$

Quiz

What's the output?



<https://godbolt.org/z/WsG4M9Tze>

```
double x = 1.0 / 3.0;  
std::print("{} ", x);
```



<https://godbolt.org/z/cTjoq3qfa>

```
float x = 1.f / 3.f;  
std::print("{} ", x);
```

Options:

(a) 0.3

☒ (b) 0.3333333333333333**3**

(c) 0.3333333333333333**4**

(d) One third

(e) *None of the above*

(a) 0.3

(b) 0.3333333**3**

(c) 0.3333333**4**

(d) One third

(e) *None of the above*

Quiz

What's the output?



<https://godbolt.org/z/WsG4M9Tze>

```
double x = 1.0 / 3.0;  
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Quiz

What's the output?



<https://godbolt.org/z/WsG4M9Tze>

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double x = 1.0 / 3.0;  
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```



<https://godbolt.org/z/cTjoq3qfa>

```
float x = 1.f / 3.f;  
std::print("{} ", x);
```



<https://godbolt.org/z/Kf99boffG>

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(a) 0.3

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(c) 0.3333333333333333**4**

(d) One third

(e) *None of the above*

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☒ (c) 0.3333333**4**

(d) One third

(e) *None of the above*



<https://godbolt.org/z/1hM48xza9>



<https://godbolt.org/z/nYbYnY6cc>



<https://godbolt.org/z/3aWxPqdfq>

$\frac{1}{3}$

0.333,333,333,333,333,333,333,333,3



$\frac{1}{3}$

$$0.333,333,333,333,333,333,333,333,333,3 = 2^{-25} \times (11,184,810 + \frac{2}{3})$$



$\frac{1}{3}$

$$0.333,333,343,267,440,795,898,437,5 = 2^{-25} \times 11,184,811$$

$$0.333,333,333,333,333,333,333,333,3 = 2^{-25} \times (11,184,810 + \frac{2}{3})$$



$$0.333,333,313,465,118,408,203,125,0 = 2^{-25} \times 11,184,810$$

$\frac{1}{3}$

— $0.333,333,343,267,440,795,898,437,5 = 2^{-25} \times 11,184,811$



— $0.333,333,333,333,333,333,333,333,3 = 2^{-25} \times (11,184,810 + \frac{2}{3})$



— $0.333,333,313,465,118,408,203,125,0 = 2^{-25} \times 11,184,810$

$\frac{1}{3}$

0.333,333,343,267,440,795,898,437,5 = $2^{-25} \times 11,184,811$



0.333,333,333,333,333,3 = $2^{-25} \times (11,184,810 + \frac{2}{3})$



0.333,333,313,465,118,408,203,125,0 = $2^{-25} \times 11,184,810$

$\frac{1}{3}$

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$\frac{1}{3}$

0.333,333,343,267,440,795,898,437,5 = $2^{-25} \times 11,184,811$

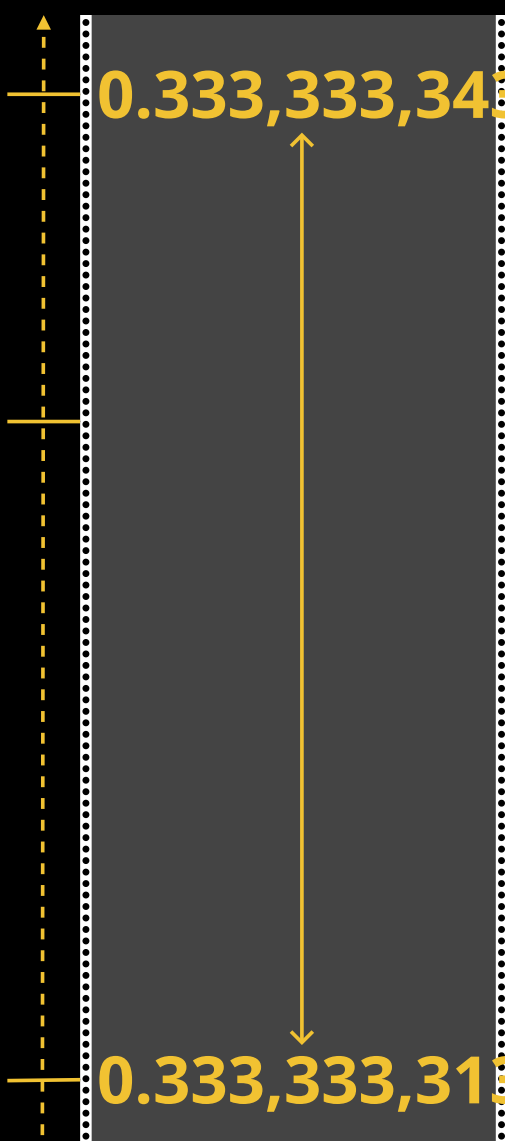


0.333,333,333,333,333,3 = $2^{-25} \times (11,184,810 + \frac{2}{3})$



0.333,333,313,465,118,408,203,125,0 = $2^{-25} \times 11,184,810$

$\frac{1}{3}$


$$0.333,333,343,267,440,795,898,437,5 = 2^{-25} \times 11,184,811$$



$$0.333,333,333,333,333,3 = 2^{-25} \times (11,184,810 + \frac{2}{3})$$



$$0.333,333,313,465,118,408,203,125,0 = 2^{-25} \times 11,184,810$$

1/3

0.333,333,343,267,440,795,898,437,5 = $2^{-25} \times 11,184,811$



0.333,333,333,333,333,3 = $2^{-25} \times (11,184,810 + \frac{2}{3})$



0.333,333,313,465,118,408,203,125,0 = $2^{-25} \times 11,184,810$

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0.333,333,343,267,440,795,898,437,5 = $2^{-25} \times 11,184,811$



0.333,333,333,333,333,3 = $2^{-25} \times (11,184,810 + \frac{2}{3})$



0.333,333,313,465,118,408,203,125,0 = $2^{-25} \times 11,184,810$

$\frac{1}{3}$

0.333,333,343,267,440,795,898,437,5 = $2^{-25} \times 11,184,811$



3,333,333,333,333,333,3 = $2^{-25} \times (11,184,810 + \frac{2}{3})$



0.333,333,313,465,118,408,203,125,0 = $2^{-25} \times 11,184,810$

$\frac{1}{3}$

0.333,333,343,267,440,795,898,437,5 = $2^{-25} \times 11,184,811$

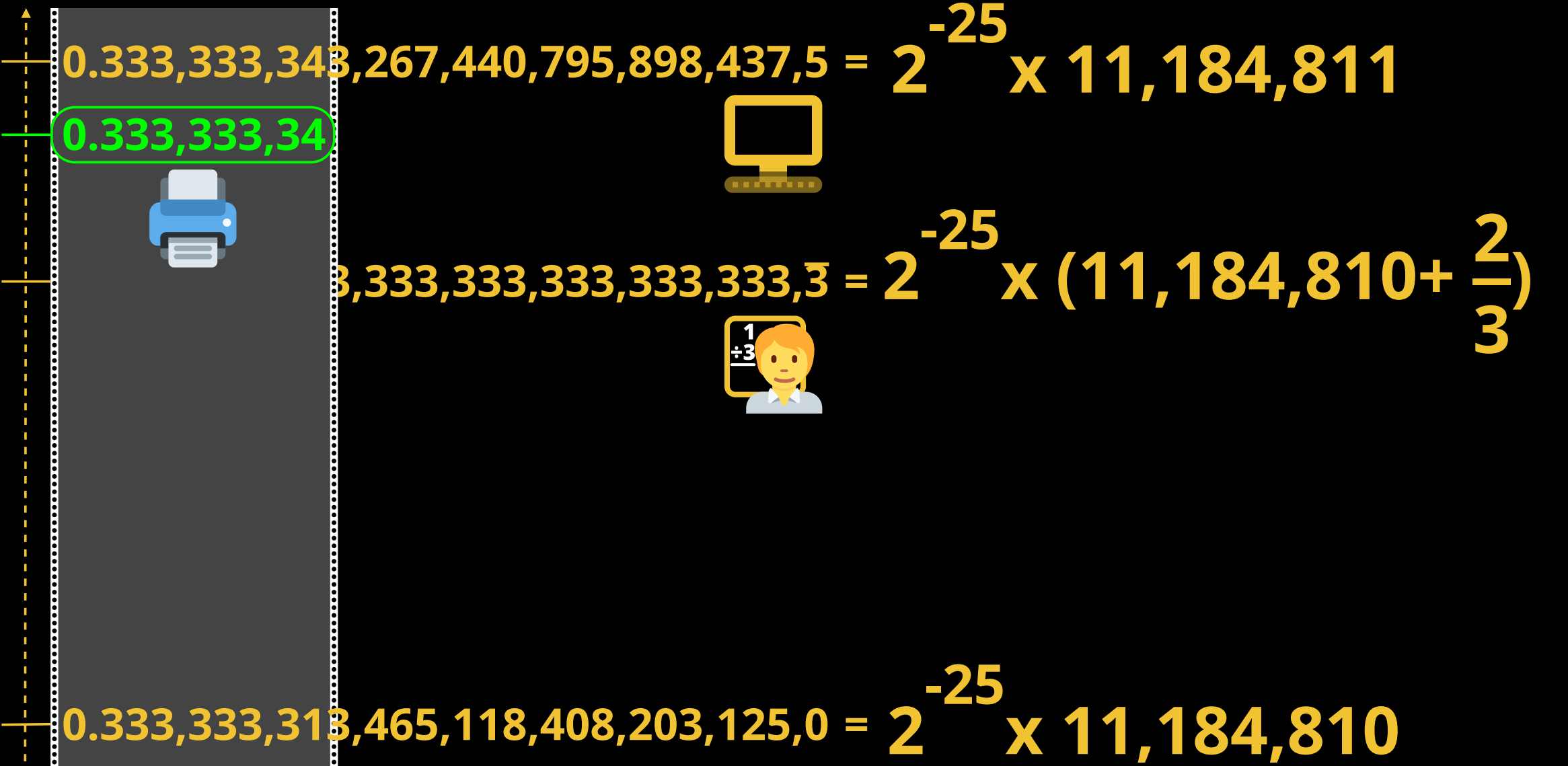


0.333,333,333,333,333,3 = $2^{-25} \times (11,184,810 + \frac{2}{3})$



0.333,333,313,465,118,408,203,125,0 = $2^{-25} \times 11,184,810$

1/3



Jorge Luis Borges, The Book of Imaginary Beings



*"There is something in the dragon's image
that fits man's imagination."*

Dragon's Den



Copyright: BBC

The dragon's den

1990 - <i>Dragon</i>	Guy L. Steele, Jon L. White
1996	Robert G. Burger and R. Kent Dybvig
2010 - <i>Grisù</i>	Florian Loitsch
2013	Aubrey Jaffer
2016 - <i>Errol</i>	Marc Andryscio, Ranjit Jhala, Sorin Lerner
2018 - <i>Ryū</i>	Ulf Adams
2020 - <i>Schubfach</i>	Raffaello Giuliatti
2020 - <i>Grisù-Exact</i>	Junekey Jeon
2022 - <i>Dragonbox</i>	Junekey Jeon

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2020 - *Schubfach* Raffaello Giuliatti

2020 - *Grisù-Exact* Junekey Jeon

2022 - *Dragonbox* Junekey Jeon



Three steps

Binary32

IEEE 754 binary32 (float) layout

biased exponent	mantissa
8	23

0 1 0 0 0 0 0 1 0 0 1 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

$+2^{-00010100} \times 10101011000000000000000000000000$

$+2^{(-20)} \times 1\ 1,2\ 7\ 2,1\ 9\ 2\ 0\ 0\ 0,0\ 0$

decode representation

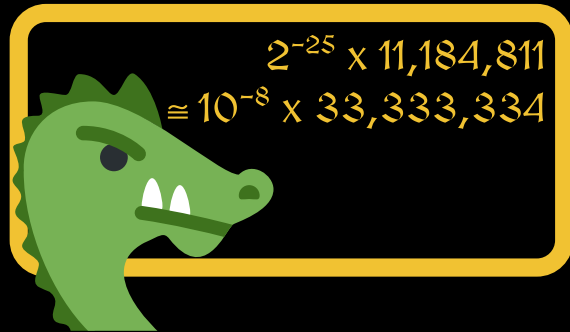
Three steps

Binary32

±	biased exponent	mantissa
1	8	23
0	1 0 0 0 0 0 1 0	0 1 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

$+2^{-00010100} \times 10101110000000000000000000$

$+2^{(-20)} \times 1\ 1,2\ 7\ 2,1\ 9\ 2\ 0\ 0\ 0,0\ 0$

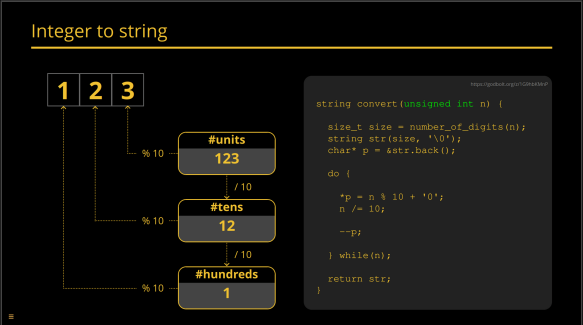
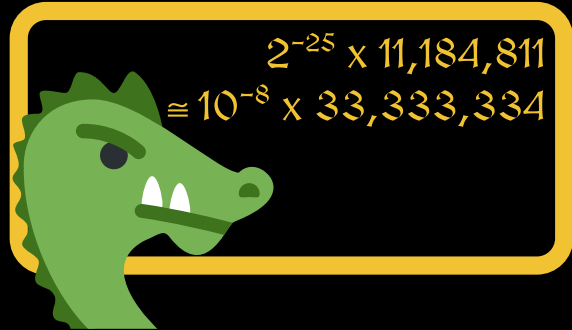
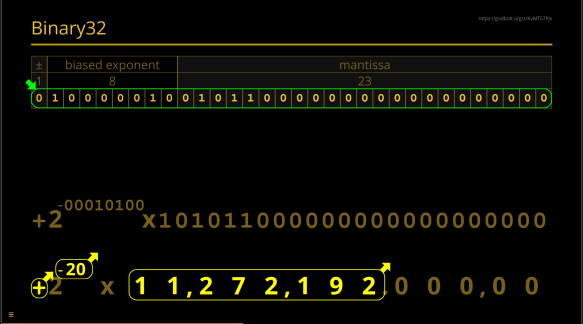


decode representation



convert binary to decimal

Three steps



decode representation



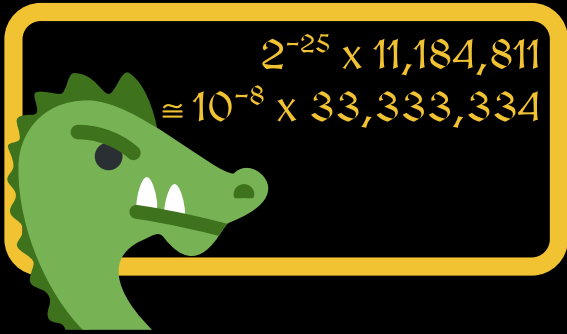
convert binary to decimal



convert exponent and mantissa to string

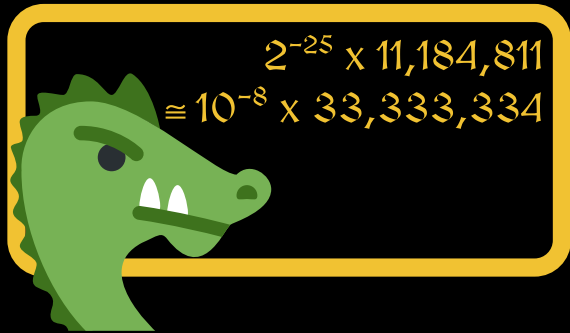
Dragon's problem

Given $m \in \mathbb{N}$ and $E \in \mathbb{Z}$,
find $n \in \mathbb{N}$ and $F \in \mathbb{Z}$ such that
$$n \times 10^F \cong m \times 2^E.$$

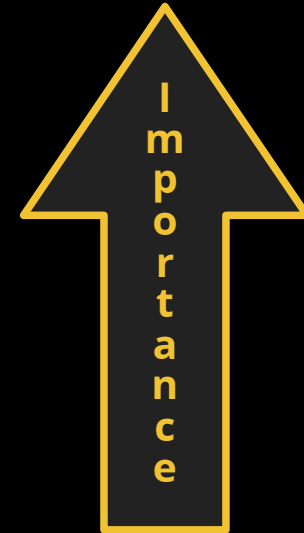


Dragon's problem

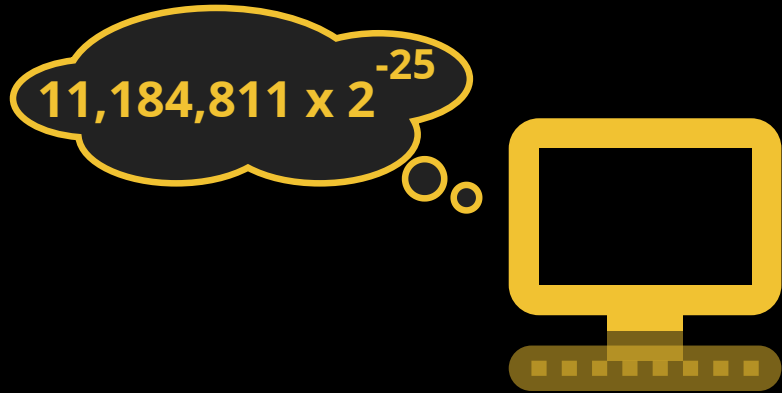
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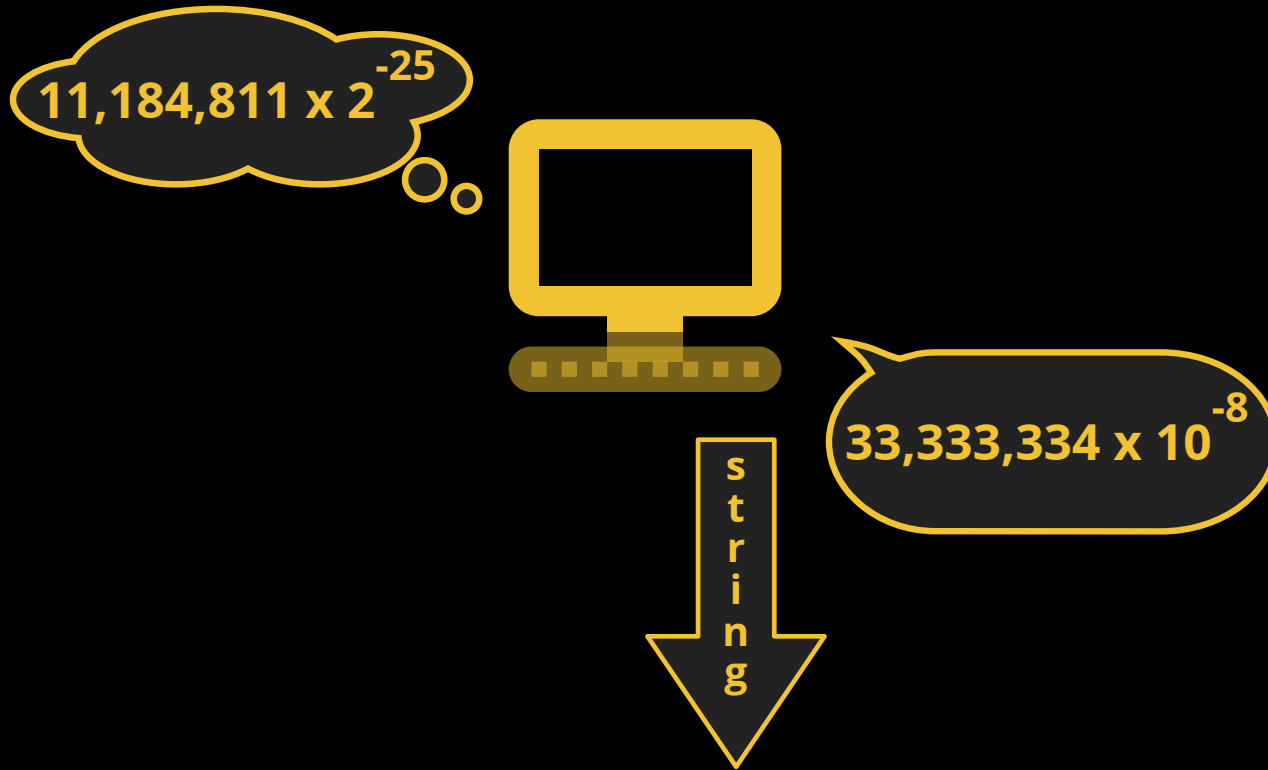
- No information loss
- As short as possible
- As close as possible
- Tiebreak rules



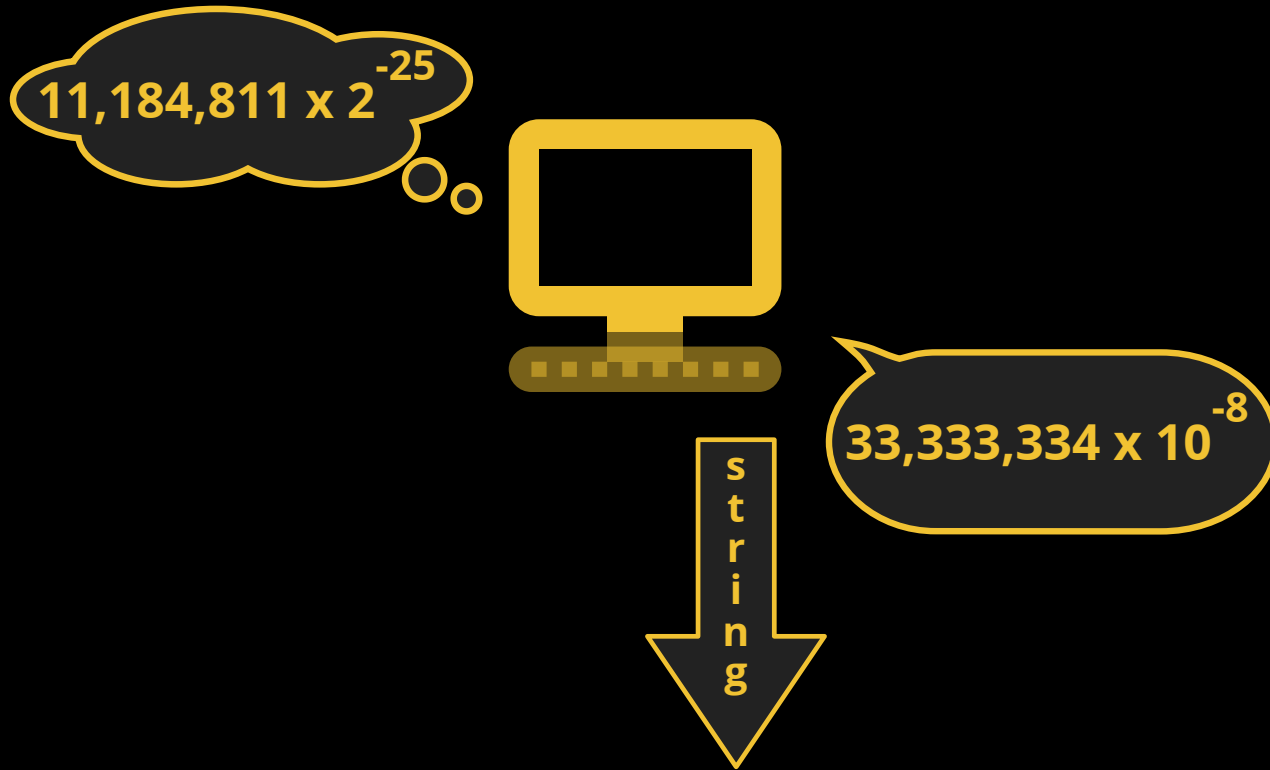
No information loss



No information loss

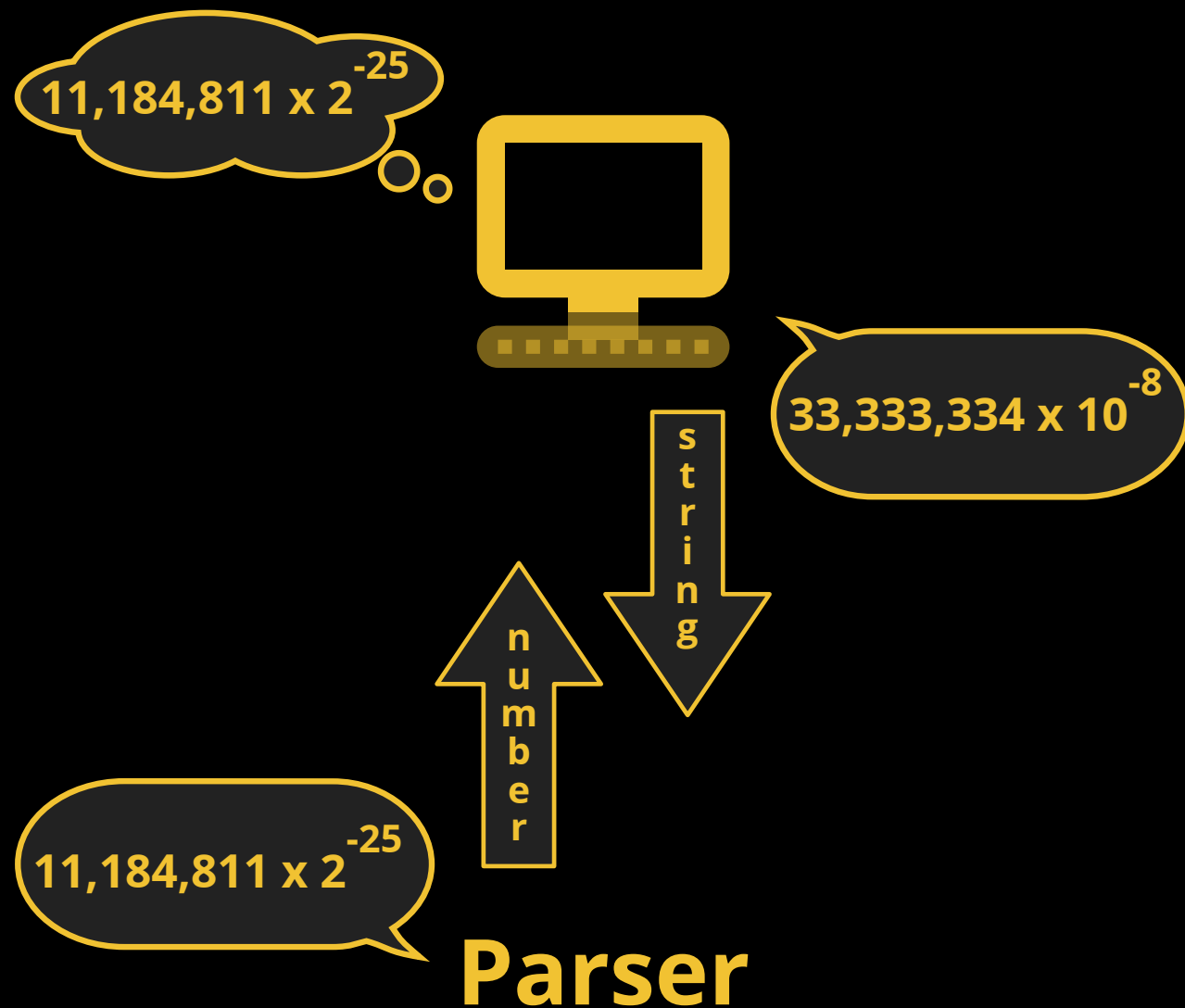


No information loss

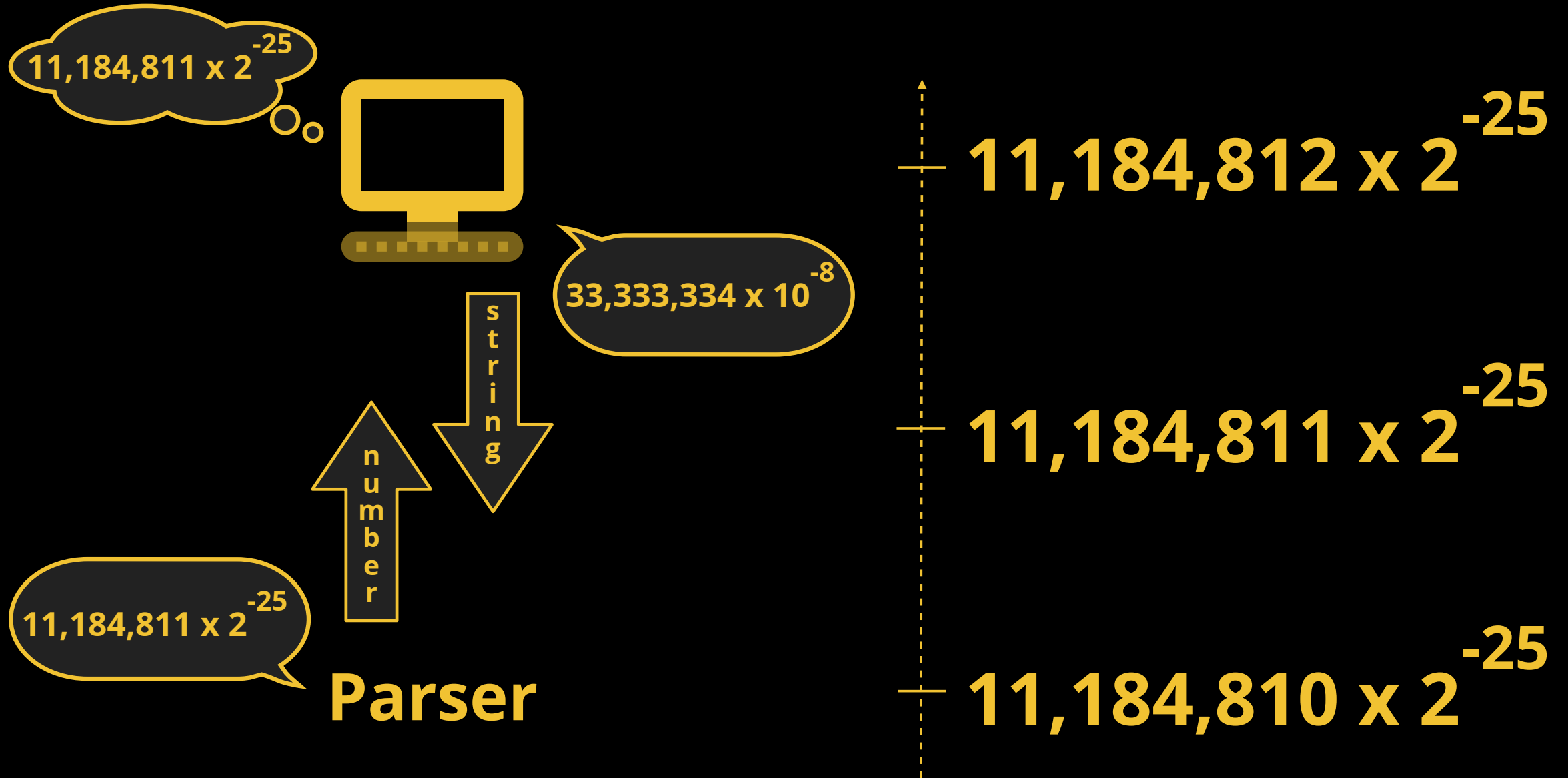


Parser

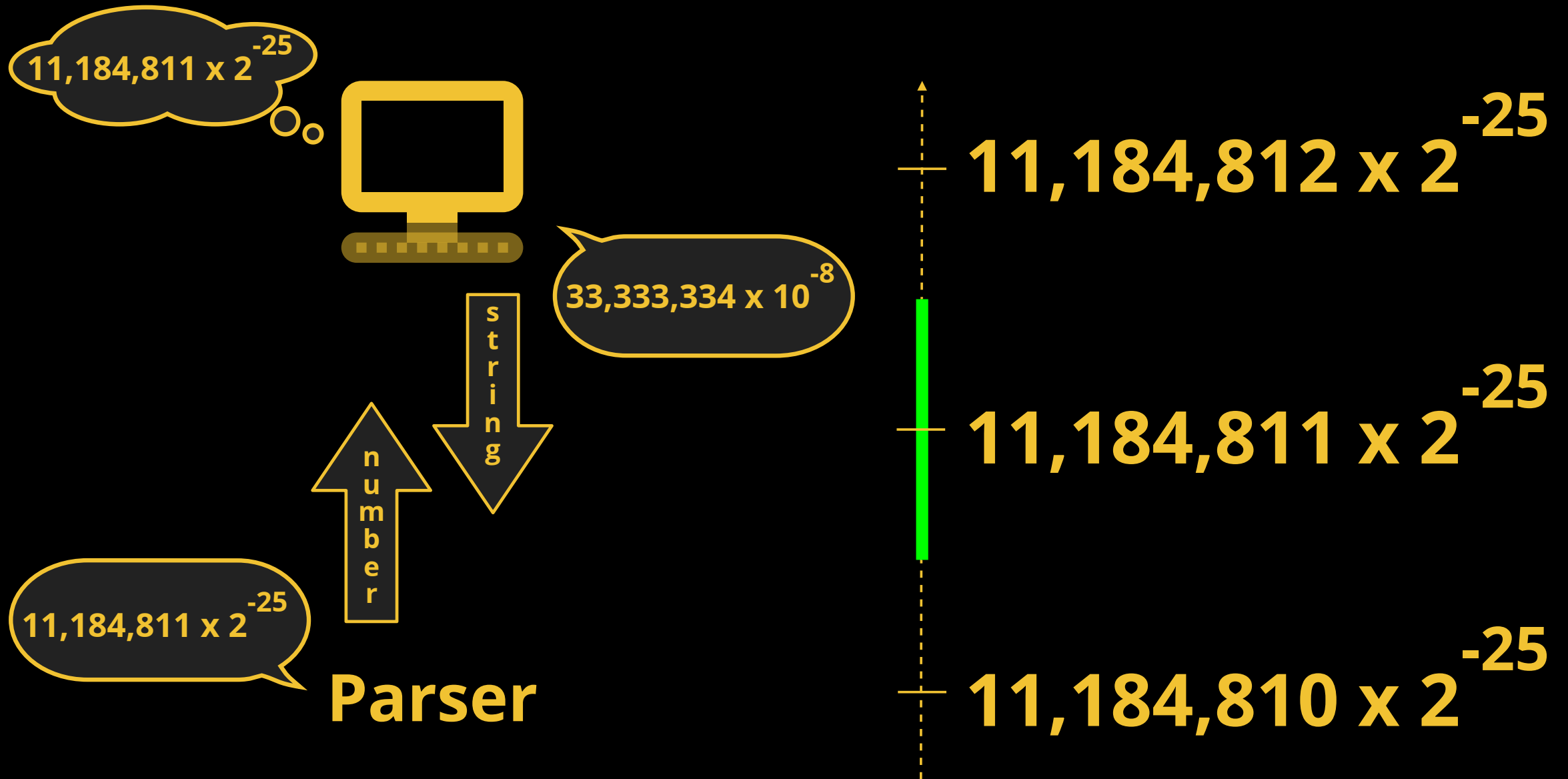
No information loss



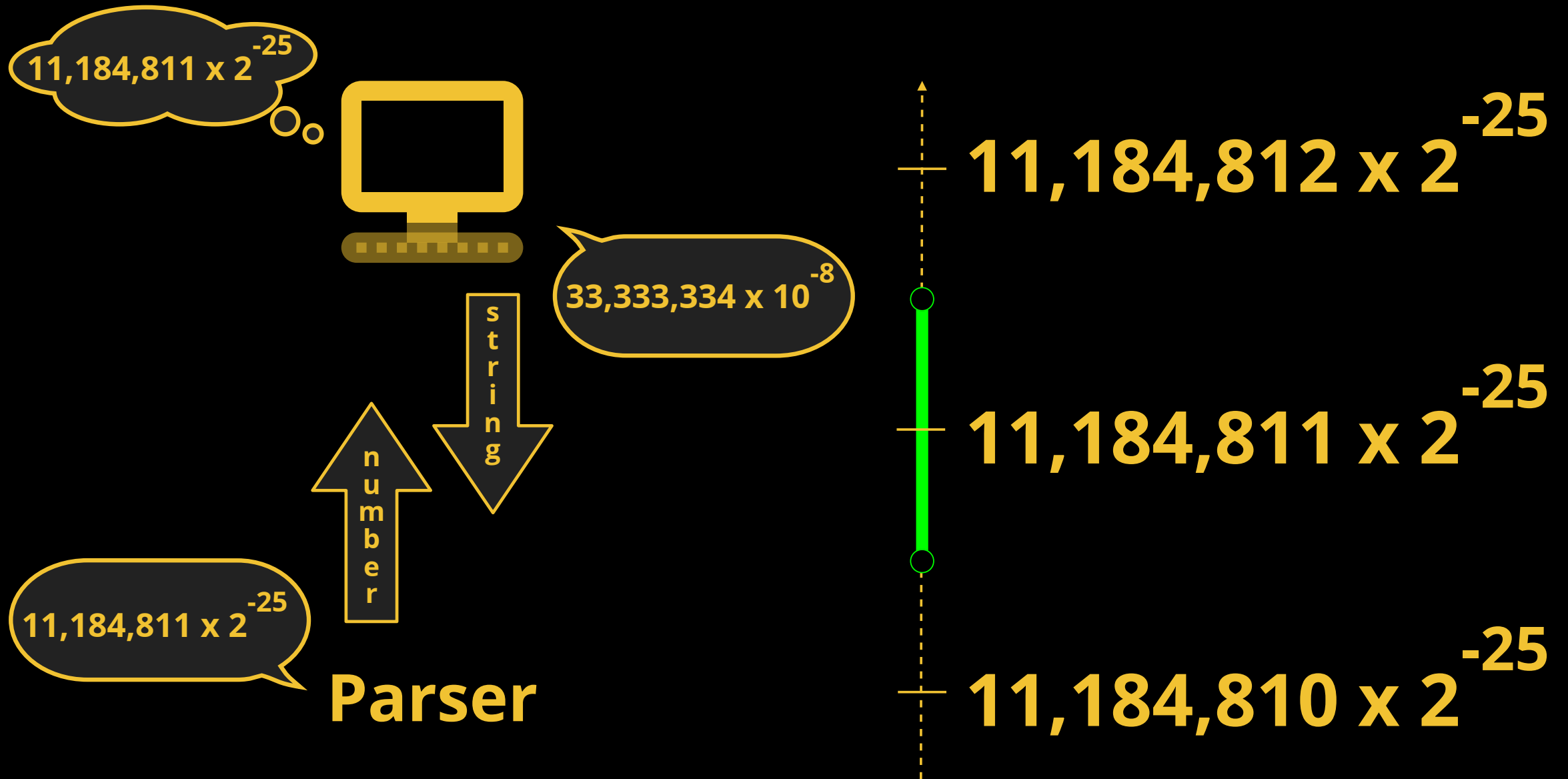
No information loss



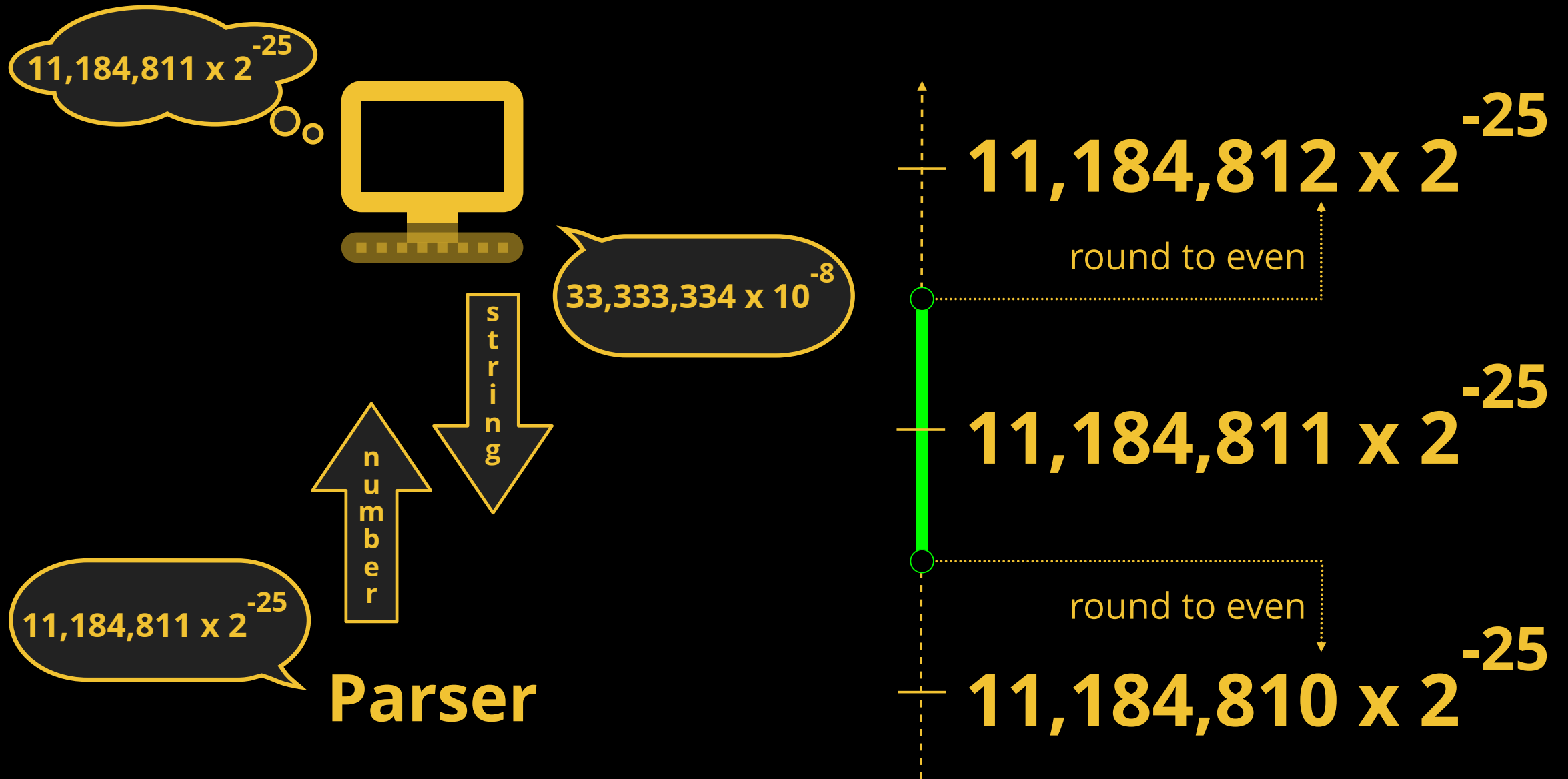
No information loss



No information loss



No information loss



As short as possible

$$9,765,625 \times 2^{10} = 10 \times 10^9$$
$$= 1 \times 10^{10}$$

As short as possible


$$9,765,625 \times 2^{10} = 10 \times 10^9$$


$$= 1 \times 10^{10}$$

Only the mantissa matters.

As short as possible

$$13,421,773 \times 2^{-27} \approx 100 \times 10^{-3}$$

$$\approx 10 \times 10^{-2}$$

$$\approx 1 \times 10^{-1}$$

As short as possible

$$13,421,773 \times 2^{-27} \approx 100 \times 10^{-3}$$

$$\approx 10 \times 10^{-2}$$

$$\approx 1 \times 10^{-1}$$

$$\approx 0.1$$

As short as possible

$$13,421,773 \times 2^{-27} \approx 100 \times 10^{-3}$$

$$\approx 10 \times 10^{-2}$$

only representations of
the form $n \times 10^F$ matter

$$\approx 1 \times 10^{-1}$$

$$\approx 0.1$$



As short as possible

$$13,421,773 \times 2^{-27} \approx 100 \times 10^{-3}$$

$$\approx 10 \times 10^{-2}$$

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$\approx$



As short as possible



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only representations of
the form $n \times 10^F$ matter

$$\approx 1 \times 10^{-1}$$

the larger F ,
the shorter n

$\approx$



As short as possible

$$10,066,330 \times 2^{-25}$$



$$30,000,002 \times 10^{-8}$$

$$30,000,001 \times 10^{-8}$$

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As short as possible

$$10,066,330 \times 2^{-25}$$



$$30,000,002 \times 10^{-8}$$

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mantissa is multiple of 10
 $\implies$ we can do better

$$3 \times 10^{-1}$$

As short as possible


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mantissa is multiple of 10
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$$3 \times 10^{-1}$$

Pythagoras



*“Leave the roads;
take the trails.”*

Guaraní people



Guaraní mythology

<https://tinyurl.com/75xhk7ef>



Guaraní mythology

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1 God **Tupã** and goddess **Arasy** created the first humans, **Rupave** and **Sypave**.



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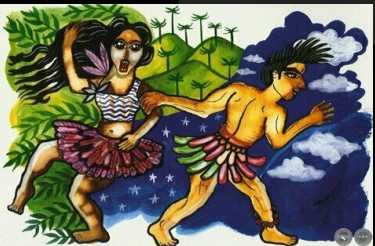
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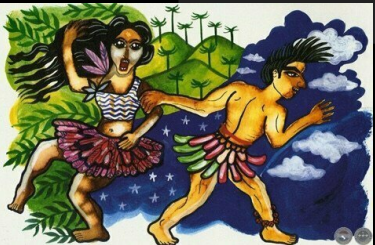
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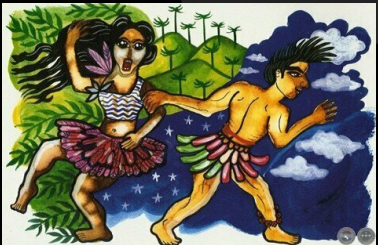
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7 **Tejú Jaguá** was the first legendary monster.

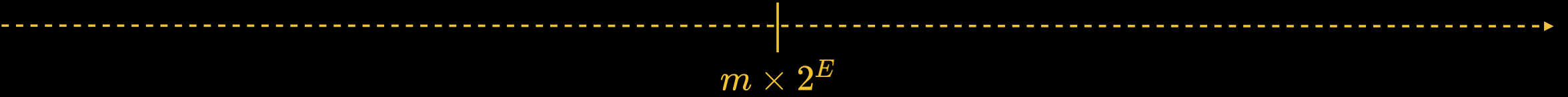


Tejú Jaguá - I

$$n \times 10^F \cong m \times 2^E$$

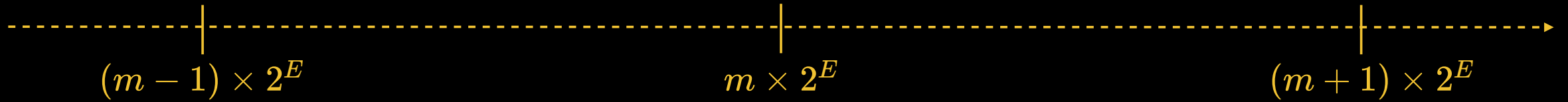
Tejú Jaguá - I

$$n \times 10^{\textcircled{F}} \cong m \times 2^E$$


$$m \times 2^E$$

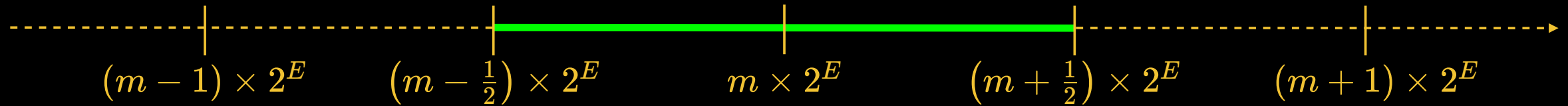
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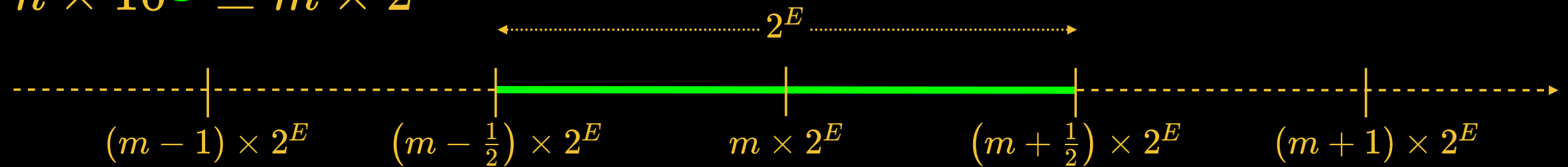
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Tejú Jaguá - I

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Tejú Jaguá - I

$$n \times 10^{\textcircled{F}} \cong m \times 2^E$$

$$\xleftrightarrow{2^E}$$



Tejú Jaguá - I

$$n \times 10^{\textcircled{F}} \cong m \times 2^E$$

$$\xleftarrow{\hspace{1.5cm} 2^E \hspace{1.5cm} \xrightarrow{\hspace{1.5cm}}}$$



- The larger F , the shorter n .

Tejú Jaguá - I

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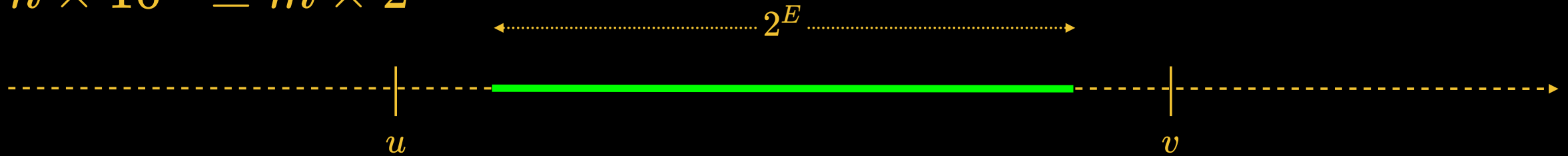
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- The larger F , the shorter n .
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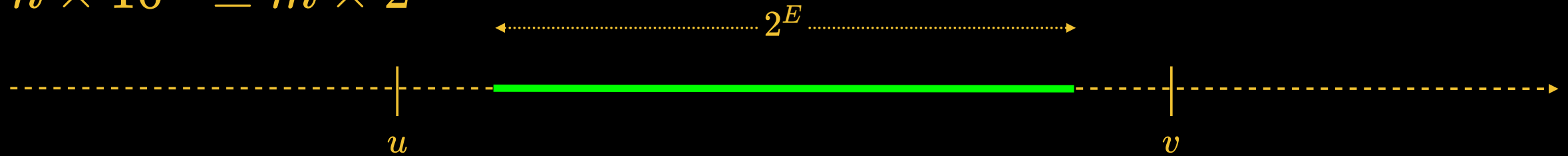
$$n \times 10^{\textcircled{F}} \cong m \times 2^E$$



- The larger F , the shorter n .
- We wish to ensure that $n \times 10^F$ falls inside the permissible interval.
- If we fail, then two **consecutive** numbers u and v of this form fall outside the permissible interval and thus, $10^F = v - u > 2^E$.

Tejú Jaguá - I

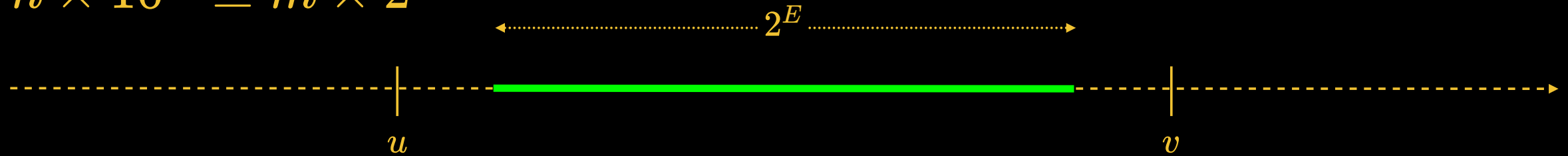
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- We set F to be the largest integer such that $10^F \leq 2^E$, i.e.,

Tejú Jaguá - I

$$n \times 10^{\textcolor{red}{F}} \cong m \times 2^E$$



- The larger F , the shorter n .
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- If we fail, then two **consecutive** numbers u and v of this form fall outside the permissible interval and thus, $10^F = v - u > 2^E$.
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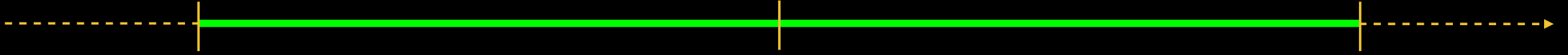
$$10^F \leq 2^E < 10^{F+1} \iff F \leq E \times \log(2) < F + 1 \iff \textcolor{red}{F = \lfloor E \times \log(2) \rfloor}$$

Tejú Jaguá - II

$$\left(m - \frac{1}{2}\right) \times 2^E$$

$$m \times 2^E$$

$$\left(m + \frac{1}{2}\right) \times 2^E$$

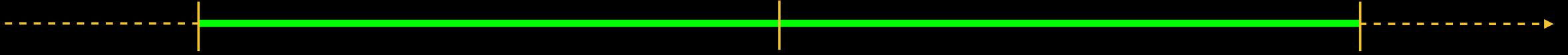


Tejú Jaguá - II

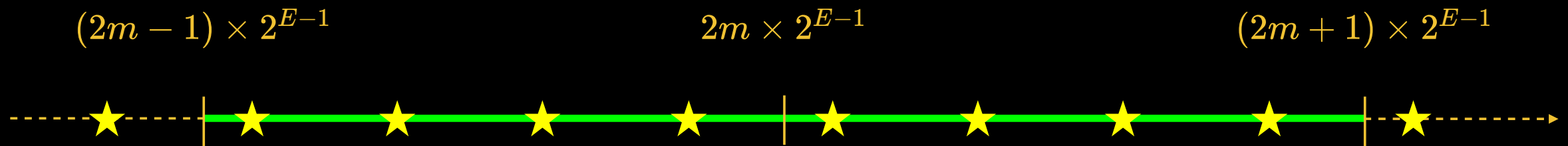
$$(2m - 1) \times 2^{E-1}$$

$$2m \times 2^{E-1}$$

$$(2m + 1) \times 2^{E-1}$$

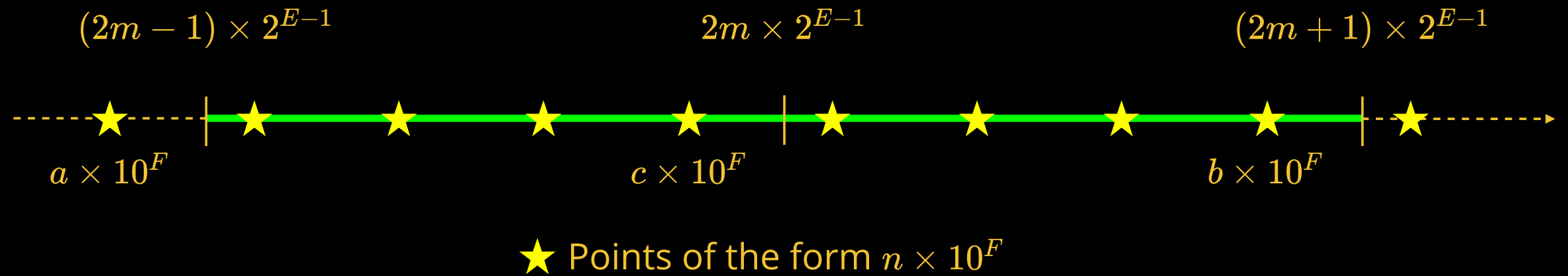


Tejú Jaguá - II

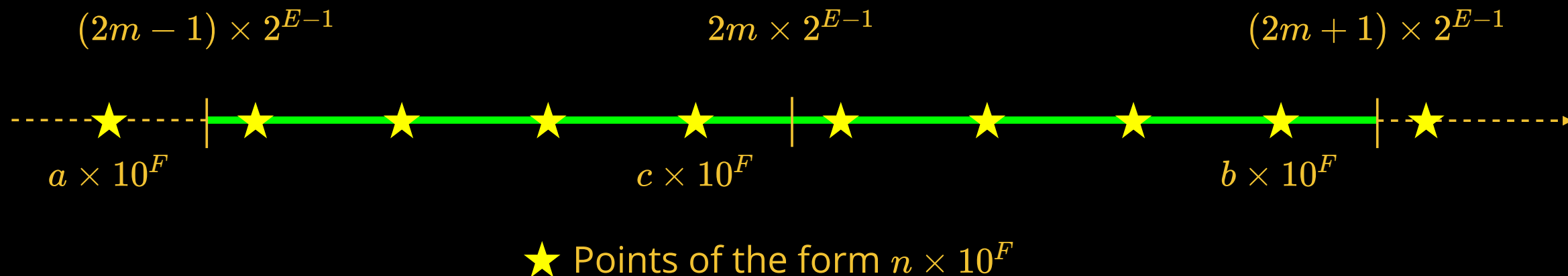


★ Points of the form $n \times 10^F$

Tejú Jaguá - II

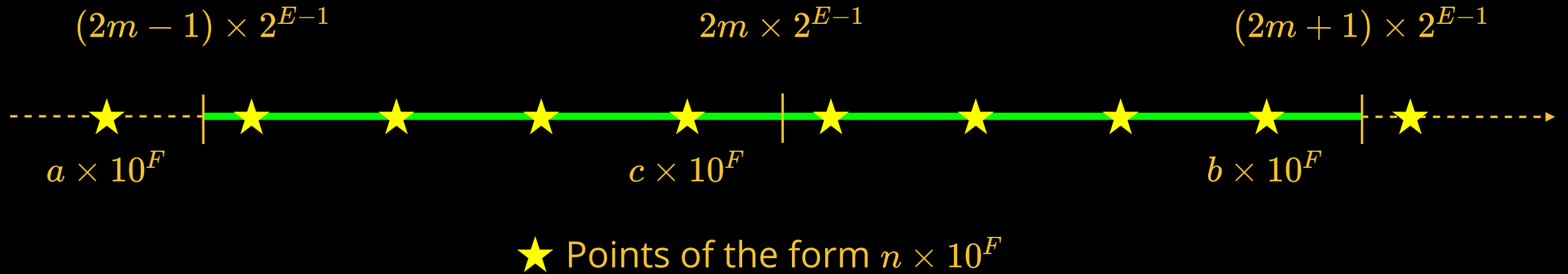


Tejú Jaguá - II



$$a \times 10^F \leq (2m-1) \times 2^{E-1} < (a+1) \times 10^F$$

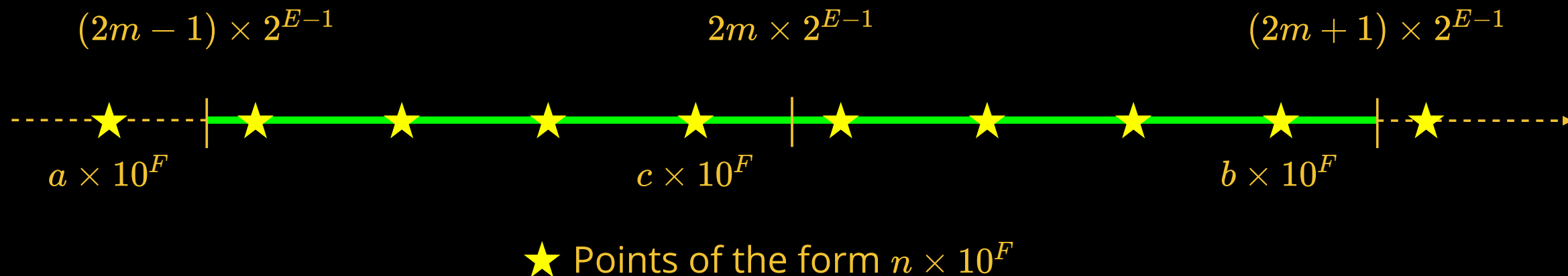
Tejú Jaguá - II



$$a \times 10^F \leq (2m - 1) \times 2^{E-1} < (a + 1) \times 10^F$$

$$a \leq (2m - 1) \times \frac{2^{E-1}}{10^F} < a + 1$$

Tejú Jaguá - II

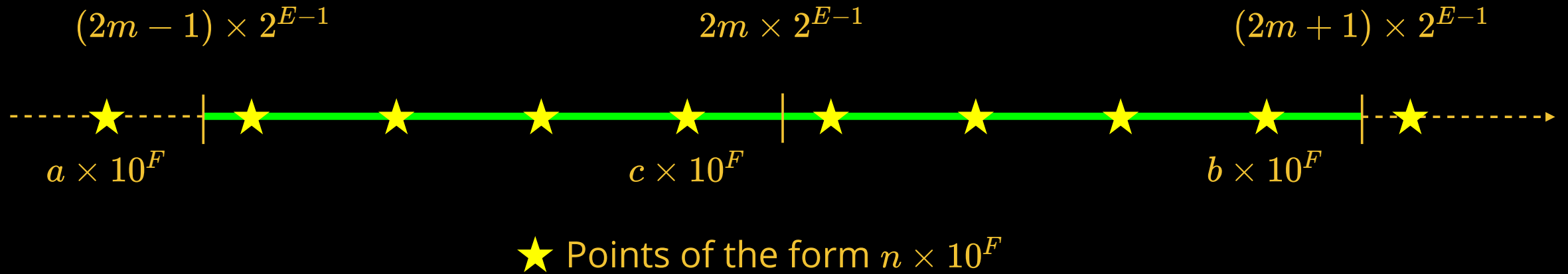


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$$a \leq (2m - 1) \times \frac{2^{E-1}}{10^F} < a + 1$$

$$a = \left\lfloor (2m - 1) \times \frac{2^{E-1}}{10^F} \right\rfloor$$

Tejú Jaguá - II



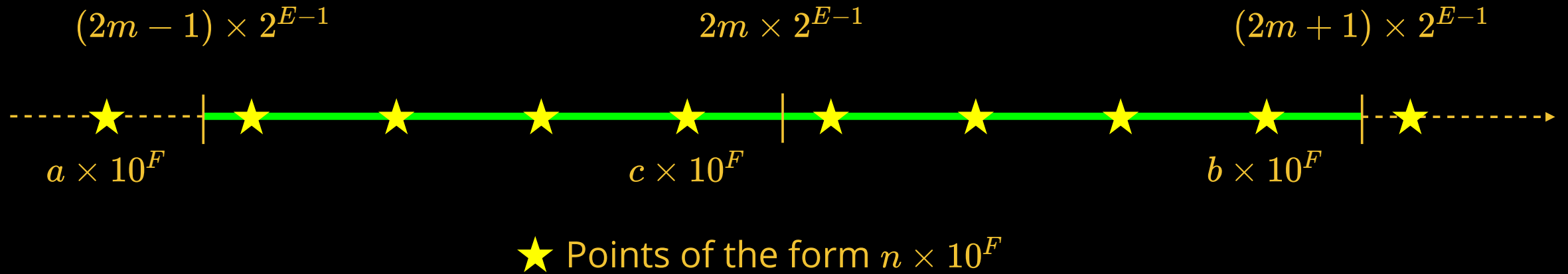
$$a \times 10^F \leq (2m - 1) \times 2^{E-1} < (a + 1) \times 10^F$$

$$b = \left\lfloor (2m + 1) \times \frac{2^{E-1}}{10^F} \right\rfloor$$

$$a \leq (2m - 1) \times \frac{2^{E-1}}{10^F} < a + 1$$

$$a = \left\lfloor (2m - 1) \times \frac{2^{E-1}}{10^F} \right\rfloor$$

Tejú Jaguá - II



$$a \times 10^F \leq (2m - 1) \times 2^{E-1} < (a + 1) \times 10^F$$

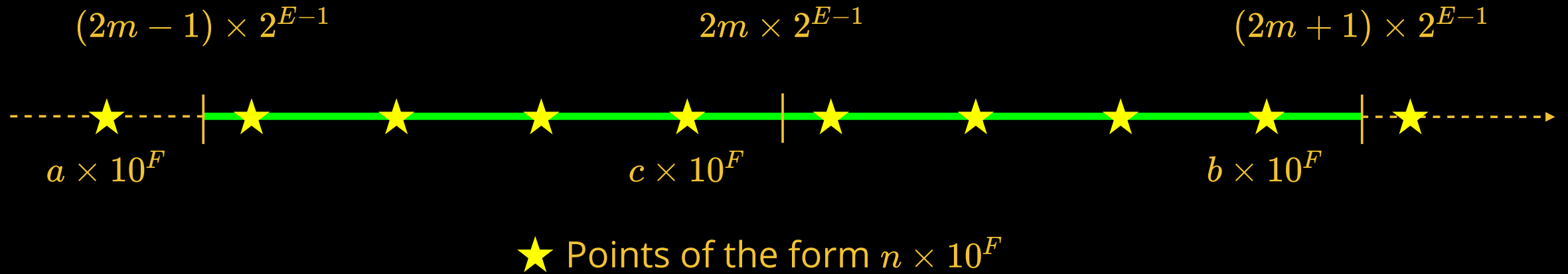
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$$a = \left\lfloor (2m - 1) \times \frac{2^{E-1}}{10^F} \right\rfloor$$

$$c = \left\lfloor 2m \times \frac{2^{E-1}}{10^F} \right\rfloor$$

Tejú Jaguá - II



$$a \times 10^F \leq (2m - 1) \times 2^{E-1} < (a + 1) \times 10^F$$

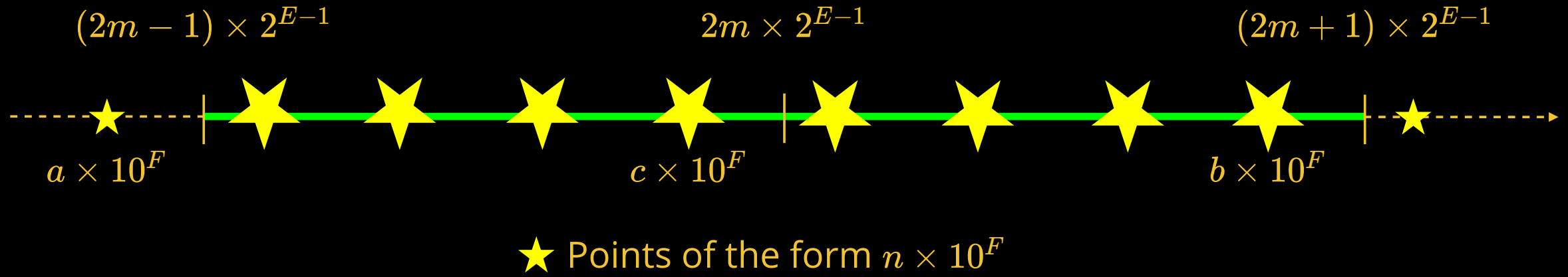
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$$a = \left\lfloor (2m - 1) \times \frac{2^{E-1}}{10^F} \right\rfloor$$

$$b = \left\lfloor (2m + 1) \times \frac{2^{E-1}}{10^F} \right\rfloor$$

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Tejú Jaguá - III

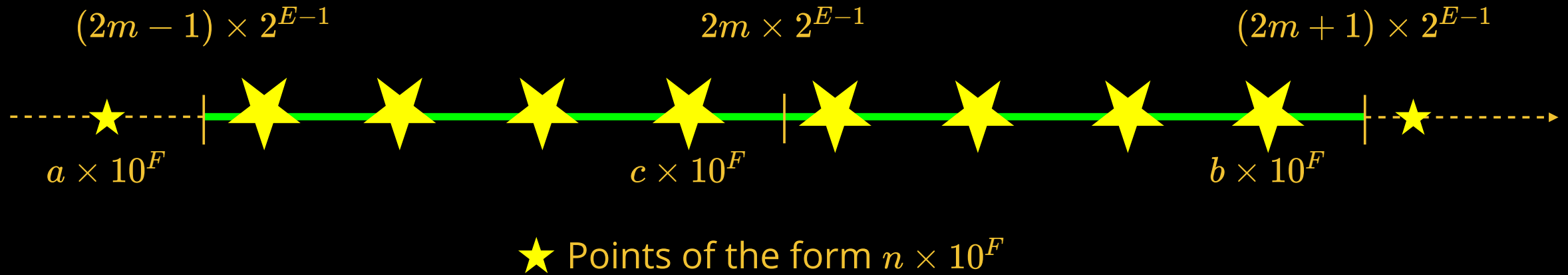


Is any of those ★ of the form $s \times 10^F$ where s is a multiple of 10?



If so, then we can do better.

Tejú Jaguá - III



Is any of those ★ of the form $s \times 10^F$ where s is a multiple of 10?



If so, then we can do better.

A good **candidate** is $s = 10 \left\lfloor \frac{b}{10} \right\rfloor$.

Return

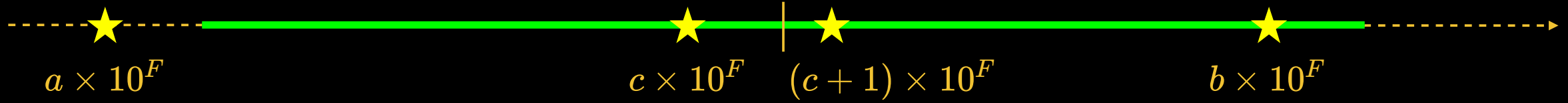
- the shortest^(*), $s \times 10^F$, if you **can**; or
- the closest, $c \times 10^F$ or $(c + 1) \times 10^F$, if you **must**.

(*) After removing trailing zeros and adjusting exponent.

Tejú Jaguá - V

$$\odot s \times 10^F$$

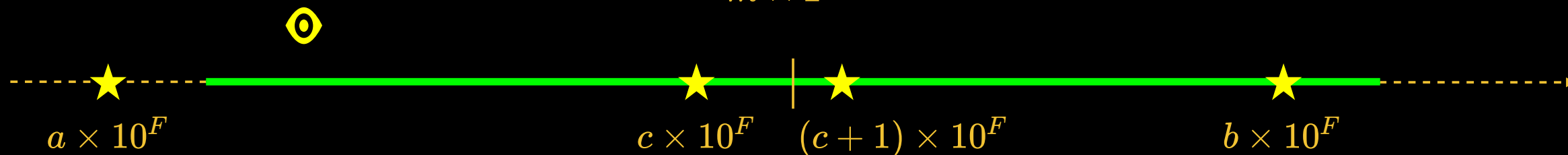
$$m \times 2^E$$



Tejú Jaguá - V

$$\odot s \times 10^F$$

$$m \times 2^E$$



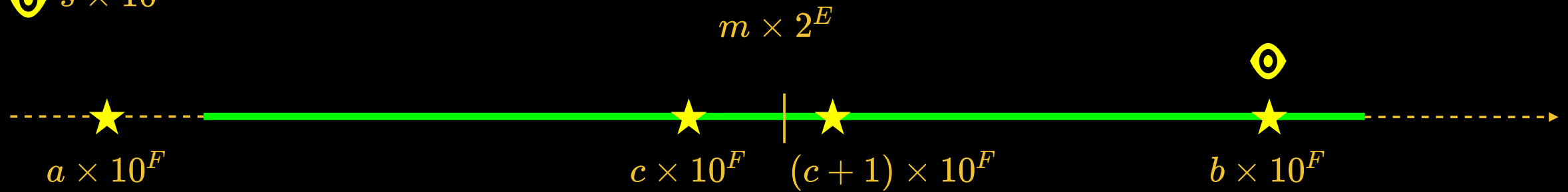
if $a < s$

if $s < b$

return the shortest, $s \times 10^F$

Tejú Jaguá - V

⊙ $s \times 10^F$



if $a < s$

if $s < b$

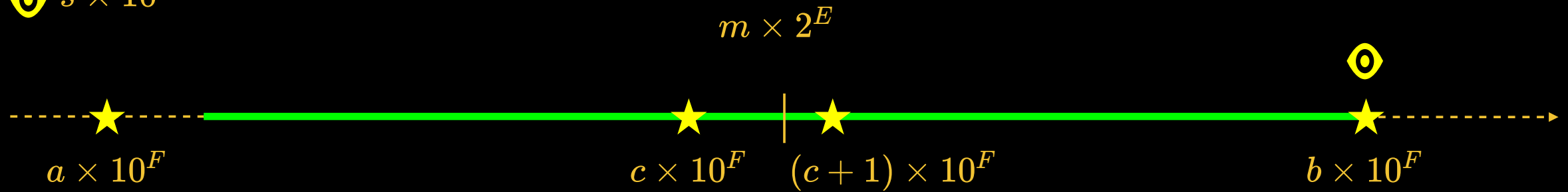
return the shortest, $s \times 10^F$

if b is not a tie

return the shortest, $s \times 10^F$

Tejú Jaguá - V

⊙ $s \times 10^F$



if $a < s$

if $s < b$

return the shortest, $s \times 10^F$

if b is not a tie

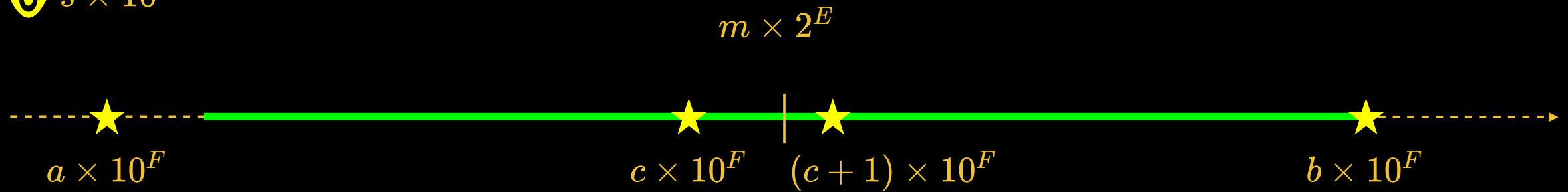
return the shortest, $s \times 10^F$

if m wins against $m+1$

return the shortest, $s \times 10^F$

Tejú Jaguá - V

① $s \times 10^F$



if $a < s$

if $s < b$

return the shortest, $s \times 10^F$

if b is not a tie

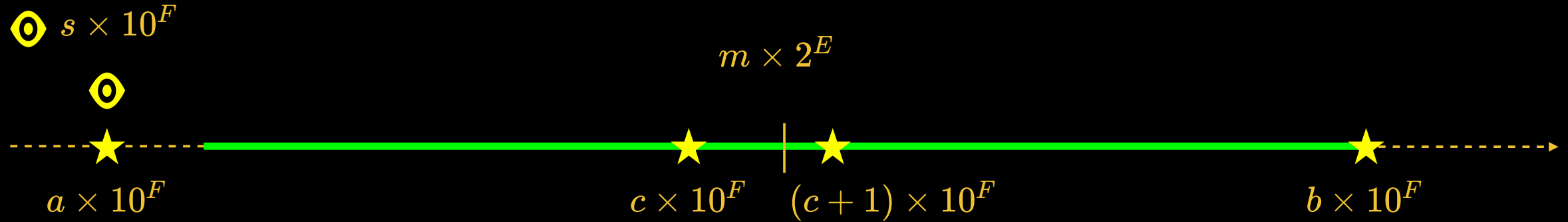
return the shortest, $s \times 10^F$

if m wins against $m + 1$

return the shortest, $s \times 10^F$

return the closest, $c \times 10^F$ or $(c + 1) \times 10^F$

Tejú Jaguá - V



if $a < s$

if $s = a$

if $s < b$

return the shortest, $s \times 10^F$

if b is not a tie

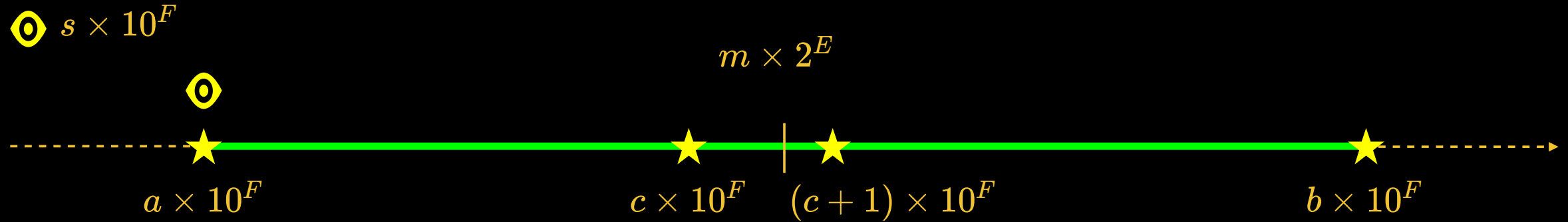
return the shortest, $s \times 10^F$

if m wins against $m+1$

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return the closest, $c \times 10^F$ or $(c+1) \times 10^F$

Tejú Jaguá - V



if $a < s$

if $s < b$

return the shortest, $s \times 10^F$

if b is not a tie

return the shortest, $s \times 10^F$

if m wins against $m + 1$

return the shortest, $s \times 10^F$

return the closest, $c \times 10^F$ or $(c + 1) \times 10^F$

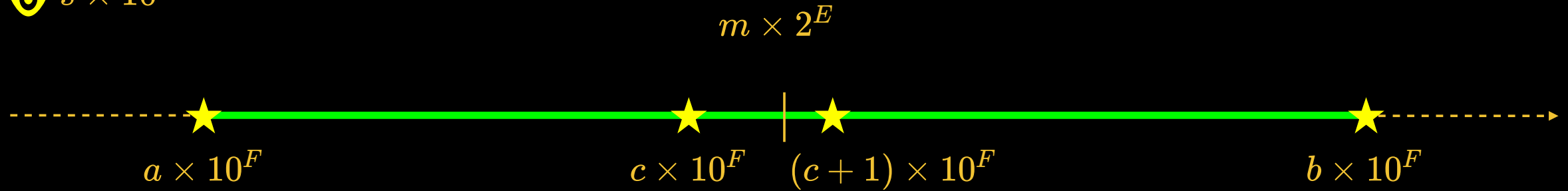
if $s = a$

if a is a tie and m wins against $m - 1$

return the shortest, $s \times 10^F$

Tejú Jaguá - V

① $s \times 10^F$



if $a < s$

if $s < b$

return the shortest, $s \times 10^F$

if b is not a tie

return the shortest, $s \times 10^F$

if m wins against $m + 1$

return the shortest, $s \times 10^F$

return the closest, $c \times 10^F$ or $(c + 1) \times 10^F$

if $s = a$

if a is a tie and m wins against $m - 1$

return the shortest, $s \times 10^F$

return the closest, $c \times 10^F$ or $(c + 1) \times 10^F$

Practicalities

$$F = \lfloor E \times \log(2) \rfloor = \lfloor E \times 0.301... \rfloor$$

$$F = \left\lfloor \frac{1,292,913,987 \times E}{2^{32}} \right\rfloor$$

Valid $\forall E \in [-112,815, 112,815]$

Practicalities

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$$F = \left\lfloor \frac{1,292,913,987 \times E}{2^{32}} \right\rfloor$$

Valid $\forall E \in [-112,815, 112,815]$

$$a = \left\lfloor M \times \frac{2^{E-1}}{10^F} \right\rfloor$$

$$a = \left\lfloor \frac{M \times M(E)}{2^K} \right\rfloor$$

K = twice the size of floating-point type

$M(E) \cong 2^K \times \frac{2^{E-1}}{10^F}$ can be precomputed
and stored in a look-up table

Practicalities

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Detecting ties becomes a divisibility check by 5^F using the *minverse* algorithm

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Optimisation for c

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Optimisation for integers

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K = twice the size of floating-point type

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Optimisation for c

Practicalities

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Detecting ties becomes a divisibility check by 5^F using the *minverse* algorithm

Optimisation for integers

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K = twice the size of floating-point type

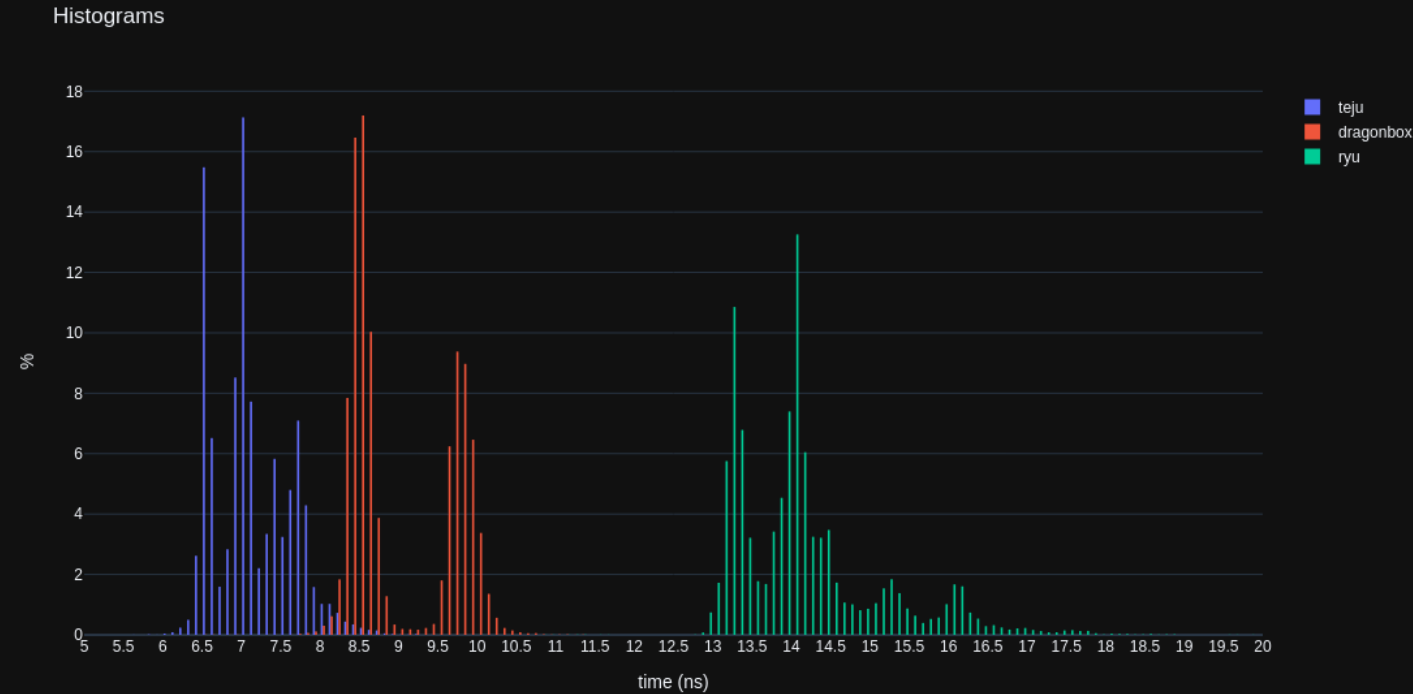
$M(E) \cong 2^K \times \frac{2^{E-1}}{10^F}$ can be precomputed and stored in a look-up table

Optimisation for c

Other tricks

Benchmarks (centred)

Intel i7 (10510U) - gcc 13.2.1



teju x dragonbox

wins 99.5%

ties 0.0%

losses 0.5%

teju x ryu

wins 100.0%

ties 0.0%

losses 0.0%

	mean	σ	min	median	max	relative
teju	7.14	0.50	2.84	7.06	9.63	1.00
dragonbox	9.04	0.66	6.52	8.64	11.39	1.27
ryu	14.16	0.98	3.40	14.01	20.50	1.98

Joseph Campbell



“The very cave you are afraid to enter turns out to be the source of what you are looking for.”

Tejú Jaguá



Tejú Jaguá



Thank you

Cristina Acosta

Aldo Galeano

Lorenz Schneider

Manuel Caicoya

Victor Bogado

